Linear Algebra (MT-1004)

Homework#06

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Question 1

For the given matrix

$$A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 0 & 1 & 3 \\ -1 & 3 & 6 & 1 \end{bmatrix}$$

- Write the general formula for transformation defined by $T(x) = Ax$. (Mention the domain and codomain of T)
- Find the transformation of $x = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 2 \end{bmatrix}$
- Find x such that $T(x) = 0$